Final project of Data Visualization
Part I: Dataset selection

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1 Justification of dataset selection

The chosen datasets — biodiversity occurrence records of plants and mammals in Ireland from the National Biodiversity Data Centre — were selected for both personal and professional reasons.

- **Personal and academic interest**: My background in Environmental Sciences and Biosciences has led to a strong interest in biodiversity, ecology and environmental impact.
- **Professional relevance**: These datasets align with the type of analysis I wish to develop in Data Science, particularly in nature conservation, ecological monitoring and spatial analysis.
- **Analytical suitability**: The data offer real, rich, heterogeneous ecological information that supports spatial visualisation, temporal exploration and biodiversity pattern analysis.

2 Analysis of context, timeliness, and gender perspective

This dataset is highly relevant for several reasons:

- **Timeliness**: Many records are recent and come from ongoing ecological monitoring efforts in Ireland.
- **Thematic significance**: Biodiversity is a global priority within environmental policy (EU Biodiversity Strategy, Agenda 2030, habitat and species conservation).
- **Stakeholder importance**: The data are useful for environmental agencies, researchers, naturalists, conservation managers and citizen science networks.
- **Gender perspective**: Although the dataset does not include gender information, the project acknowledges the value of inclusivity in science and the contributions of diverse groups involved in data collection. However, gender-based segmentation is not possible from the available data.

3 Dataset complexity

The dataset meets and exceeds the required level of complexity.

- **Number of records**:
  - **Plant dataset**: $> 30,000$ observations
  - **Mammal dataset**: $> 10,000$ observations
  - **Combined dataset**: tens of thousands of records

- **Variables available**:

17 variables, including spatial coordinates, temporal information, taxonomic data, sampling methods and record metadata.

- **Types of data included**:
  - **Categorical**: species, county, site names, sampling method, abundance, projection, type of individual
  - **Spatial**: spatial coordinates, precision
  - **Temporal**: dates and date ranges
  - **Text fields**: recorder names
  - **Boolean**: validation and abundance flags

- **Heterogeneity**:

Non-trivial, multi-source data including real spatial information, multiple taxonomic groups, presence/absence logic, diverse sampling methods and recorder information.

- **Combination of sources**:

  The project integrates datasets from different taxonomic groups and surveys, creating a richer combined dataset.

  - **Plants:** Biodiversity Ireland - Dataset 190 and Data Europa - B52CA650.
  - **Mammals:** Biodiversity Ireland - Dataset 253.

  This provides analytical depth and enables multi-dimensional visualisations.

4 Originality

The dataset choice and approach are original due to:

- **Non-standard theme**: Not based on common classroom datasets (COVID-19, crime, traffic, Kaggle classics).
- **Novel integration**: Combines plant and mammal observations into a single, enriched dataset not originally available.
- **Innovative perspective**: The project focuses on spatial-temporal dynamics, interactive mapping and species co-occurrence exploration.
- **New metrics to be generated**:
  - Biodiversity indices (Shannon, Simpson) by county/region
  - Observation density maps
  - Plant–mammal co-occurrence within spatial buffers
  - Temporal trends of species presence
- **Dataset enhancement**: It modernises and integrates ecological data typically analysed separately, allowing the study of potential cross-taxa interactions.

5 Research questions and data visualisation objectives

The defined research questions are strategically aligned with the unique capabilities of the prepared unified dataset, providing a novel framework for ecological analysis that satisfies the criteria of suitability for the chosen data, relevance to established ecological practices, and feasibility for data visualisation.

1. **Suitability for the chosen dataset**
   (a) **Spatial distribution and biodiversity levels**: These rely entirely on the East and North (numerical coordinates) and the TaxonName. The data is precise enough to allow aggregation by county or grid cell for accurate biodiversity index calculation (species richness).
   (b) **Temporal trends**: The Date variable is in the standardized date format, ensuring reliable input for time-series analysis and animation from 1837 up to the current decade.
   (c) **Species co-occurrence (causality)**: This core comparative objective is facilitated by the Type factor (Mammal/Plant), which allows for the creation of joint spatial analyses to test for ecological overlap.
   (d) **Sampling method effect**: The sampling method variable is maintained as a factor, enabling crucial quality control by comparing observation counts or richness across different survey techniques.
2. **Relevance and academic context**: The proposed questions are standard but powerful tools in ecological research:
   (a) **Mapping and hotspots**: Determining species distribution and identifying biodiversity hotspots are fundamental conservation practices. These visualisations directly inform resource management and conservation priority areas.
   (b) **Temporal analysis**: Analyzing temporal trends helps uncover the effects of long-term factors like climate change or land use change on species presence/absence over the last two centuries.
   (c) **Co-occurrence analysis (flora-fauna correlation)**: Investigating the spatial and temporal co-occurrence between plants (producers, habitat providers) and mammals (consumers) addresses a classical ecological question about habitat dependency and species interaction dynamics.
3. **Feasibility for data visualisation and research questions**:

The visualisations aim to address the following key questions:

- How are plant and mammal species distributed spatially across Ireland?
  - Interactive maps based on coordinate data.
- Which regions show the highest biodiversity levels?
  - Calculation of biodiversity indices by county or grid cell.
- Are there clear temporal trends in species presence?
  - Timelines and temporal heatmaps.
- Do plants and mammals co-occur in specific areas or time periods?
  - Spatial buffers, joint maps and species overlap analysis.
- Does the sampling method affect species detection?
  - Comparisons across different Sampling.method categories.

A detailed variable dictionary has been prepared to classify each field as either a study feature or a contextual dimension, supporting the visualisation and analytical workflow.

Variable dictionary

Table 1: Description and data types of the unified dataset variables (N=44,976)

Variable	Data Type	Description
Abundance_Simplified	Ord. Factor (5 levels)	Simplified abundance/quantity observed (e.g., Low, Medium, High). Essential for biodiversity analysis.
County	Factor (39 levels)	County name where the record was collected.
Date	Date	Date of the sighting (YYYY-MM-DD).
East	Numeric (num)	Eastern coordinate of the record location (Irish Grid system).
EndDate	Date	End date of the sampling period. Retained for temporal range calculation.
North	Numeric (num)	Northern coordinate of the record location (Irish Grid system).
Precision	Integer (int)	Precision of the geographical coordinates (in meters).
Projection	Factor (3 levels)	Coordinate system used (e.g., OSI, WGS84).
Recorder	Character (chr)	Name of the person who recorded the observation.
Sampling.method	Factor (55 levels)	Method used to collect the observation (e.g., Active sett, incidental).
SiteName	Factor (8,232 levels)	Name of the specific site or area where the record was made.
StartDate	Date	Start date of the sampling period. Used for date range and imputation.
TaxonName	Factor (776 levels)	Scientific name of the species observed (e.g., *Apodemus sylvaticus*). Key for species richness analysis.
TaxonVersionKey	Factor	Internal unique identifier for the specific taxonomic version.
Type	Factor (2 levels)	**Categorical indicator** created during merging: 'Mammal' or 'Plant'. Essential for comparative analysis.
UnderValidation	Character (chr)	Boolean flag indicating if the record is currently undergoing data quality validation ('False' or 'True').
ZeroAbundance	Character (chr)	Flag indicating whether the record reported zero abundance ('False' or 'True').